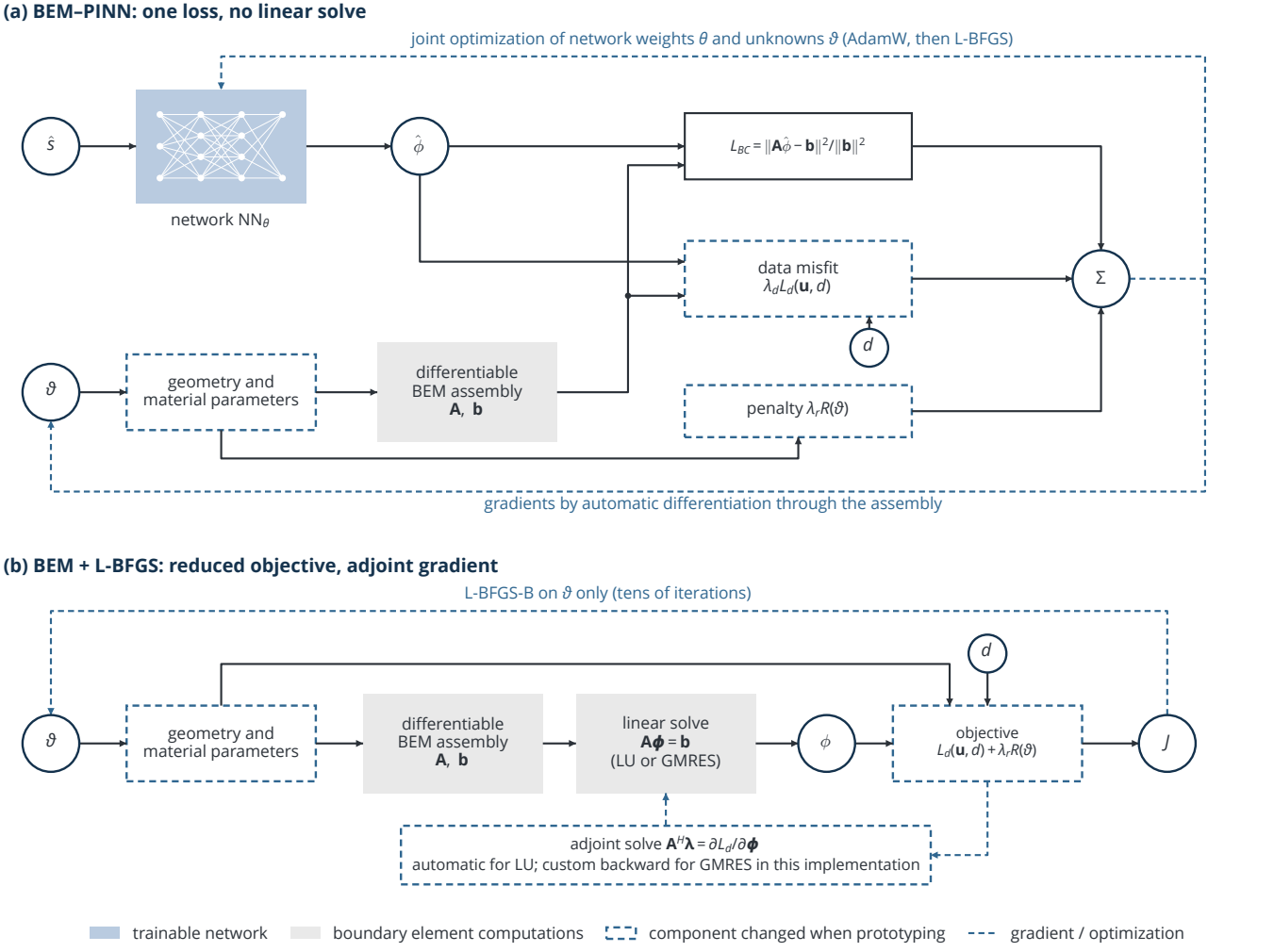
(a) BEM–PINN: one loss, no linear solve
joint optimization of network weights θ and unknowns ϑ (AdamW, then L-BFGS)
ŝ
network NN_θ
φ̂
L_BC = ||Aφ̂ − b||²/||b||²
data misfit
λ_d L_d(u, d)
d
Σ
ϑ
geometry and material parameters
differentiable BEM assembly
A, b
penalty λ_r R(ϑ)
gradients by automatic differentiation through the assembly
(b) BEM + L-BFGS: reduced objective, adjoint gradient
L-BFGS-B on ϑ only (tens of iterations)
d
ϑ
geometry and material parameters
differentiable BEM assembly
A, b
linear solve
Aφ = b
(LU or GMRES)
φ
objective
L_d(u, d) + λ_r R(ϑ)
J
adjoint solve A^H λ = ∂L_d/∂φ
automatic for LU; custom backward for GMRES in this implementation
trainable network
boundary element computations
component changed when prototyping
gradient / optimization